

# Jiwon Sung

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🌐 <https://sunione.github.io>

## EDUCATION

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### Stanford University

*Ph.D. in Electrical Engineering*

- *Advisors*: Prof. John M. Cioffi, Prof. Amin Arbabian

Sep. 2026 – Present

*Stanford, CA, USA*

### Korea Advanced Institute of Science and Technology (KAIST)

*M.S. in Electrical Engineering* | 4.12/4.3 GPA

- *Committee*: Jinseok Choi (*chair*), Junil Choi, Joonhyuk Kang
- *Dissertation*: Tracking Algorithm for Hybrid Calibration of Time-Interleaved ADCs

Feb. 2024 – Feb. 2026

*Daejeon, Korea*

### Ulsan National Institute of Science and Technology (UNIST)

*B.S. in Electrical Engineering* | 4.15/4.3 GPA (Dept. Rank: 1/51) | *Summa Cum Laude*

Feb. 2018 – Feb. 2024

*Ulsan, Korea*

## RESEARCH INTERESTS

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Chip Interconnects, Signal Processing, Wireless Communications, and Machine Learning

## RESEARCH EXPERIENCE

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### Stanford Machine Learning & Communications Lab

*Stanford University* | *Advisor*: Prof. John M. Cioffi

Mar. 2026 – Present

*Stanford, CA, USA*

### MIMO Interconnects (Joint Project with Prof. Amin Arbabian)

- TBA

### Cooperative Dynamic Spectrum Access (DSA)

- Proposed a cooperative DSA mechanism for large-scale networks that uses beamforming based on the the IEEE 1900.5.2 standard, commonly referred to as spectrum consumption models
- Simulations in an indoor warehouse setting show that the proposed DSA mechanism better utilizes the spatial dimension for spectrum sharing

### TASC Lab

*Purdue University* | *Advisor*: Prof. David J. Love

Aug. 2025 – Jan. 2026

*West Lafayette, IN, USA*

### Near-Field Velocity Sensing on an Extended Target

- Worked on near-field extended target tracking and comparing it with the point target scenario

### SIC-X LAB

*KAIST* | *Advisor*: Prof. Jinseok Choi

July 2023 – Aug. 2025

*Daejeon, Korea*

### Power-Efficient Rate-Splitting Multiple Access (RSMA) Precoding

- Developed a power-efficient and low-complexity MIMO-RSMA beamforming design that jointly optimizes the precoder, the set of active antennas, and the transmit power for a given total power budget at a base station
- Numerical results showed a new insight: medium-resolution DACs with 8 ~ 11 bits may be more power efficient than low-resolution DACs with 3 ~ 5 bits when utilizing the full potential of the available power at the base station

### Time-Interleaved ADC (TI-ADC) Mismatch Error Estimation and Compensation

- Developed a hybrid calibration algorithm for correcting offset, gain, and timing mismatch errors in TI-ADCs using the extended Kalman filter for estimation and a combination of a truncated fractional delay filter and a high-pass filter for compensation
- Our algorithm achieved 10× lower reconstruction error, did not require an FIR filter coefficient optimization, and was the first work in the literature to model time-varying mismatch errors

### InfoLab

*KAIST* | *Advisor*: Prof. Si-Hyeon Lee

Mar. 2023 – July 2023

*Daejeon, Korea*

### Local Differential Privacy (LDP) in Graph Data

- Applied LDP to decentralized graph data such that privacy is preserved while maintaining local graph statistics
- Proposed a degree-preserving asymmetric bit-flipping scheme that satisfies edge-LDP constraints

- Numerical results showed that the proposed method produced good results for coreness, a metric that quantifies users' influence in a graph
- Our method could be applied to scenarios where the task is to find potential super-spreaders in an epidemic outbreak using people's sensitive data so that the limited supply of vaccines is used on the most influential people

## Lab. of Advanced Imaging Tech.

Apr. 2022 – Aug. 2022

UNIST | Advisor: Prof. Jaejun Yoo

Ulsan, Korea

### Image-to-Image Translation

- Studied generative adversarial networks (GANs) in the context of image-to-image translation
- Tried to make improvements on *TUNIT*, a multi-domain image-to-image translation paper that uses unlabeled datasets, by eliminating the "number of domains" hyperparameter using the Incremental DBSCAN algorithm

## PUBLICATIONS

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### Journal Articles

- **Jiwon Sung**, Seokjun Park, and Jinseok Choi, "Power-Constrained and Quantized MIMO-RSMA Systems with Imperfect CSIT: Joint Precoding, Antenna Selection, and Power Control", *IEEE Transactions on Wireless Communications*, vol. 25, pp. 17634-17648, 2026
- **Jiwon Sung** and Jinseok Choi, "A New Interpretation of the Time-Interleaved ADC Mismatch Problem: A Tracking-Based Hybrid Calibration Approach", *IEEE Signal Processing Letters*, vol. 32, pp. 3710-3714, 2025

### International Conference Papers

- **Jiwon Sung**, Seungmin Choi, Abhiram R. Gorle, Atul A. Salvekar, Igor Kadota, John M. Cioffi, "Cooperative Dynamic Spectrum Access for Cell-Free MISO Networks Using Spectrum Consumption Models" (*Submitted to ICC*)
- Woocheol Shin, Taeyoung Yun, **Jiwon Sung**, Hyeongyu Kang, Jeongmin Park, Jaewoo Lee, Kyuil Sim, Dohyun Kim, Yutong He, Jinkyoo Park, "Noise-Space Reward Alignment for Few-Step Generative Models" (*Submitted to ICLR*)
- Seungmin Choi, **Jiwon Sung**, Muhammad Umer, Abhiram Rao Gorle, Guijin Son, Youngjae Yu, John M. Cioffi, "Small Models Scout Bottleneck Order for Large-Model Data Control" (*Submitted to AAAI*)
- **Jiwon Sung**, Seokjun Park, and Jinseok Choi, "Joint Optimization for Power-Constrained MIMO Systems: Is Low-Resolution DAC Still Optimal?," *2025 IEEE 101st Vehicular Technology Conference (VTC2025-Spring)*, Oslo, Norway, 2025, pp. 1-6
- Seokjun Park, **Jiwon Sung**, Jinseok Choi, Jeonghun Park, and Wonjae Shin, "Maximizing Energy and Spectral Efficiency Tradeoff in MISO-RSMA Systems Under Coarse Quantization," *2024 32nd European Signal Processing Conference (EUSIPCO)*, Lyon, France, 2024, pp. 857-861

## WORK EXPERIENCE

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### Air Force Operations Command (AFOC)

Oct. 2020 – Mar. 2022

Military Interpreter (English-Korean) | Sergeant

Pyeongtaek, Korea

## AWARDS AND SCHOLARSHIPS

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### Government-Sponsored Scholarship

Feb. 2024 – Feb. 2026

- Near-full-tuition & stipend scholarship

### National Science and Technology Scholarship

Feb. 2020 – Feb. 2024

- Full-tuition & stipend scholarship

### Semester Award

2018, 2019, 2020-Spring, 2022-Fall

- Awarded to the top students with the highest GPA scores

### Academic Performance Scholarship

Feb. 2018 – Feb. 2024

- Full-tuition scholarship

## TEACHING AND MENTORING EXPERIENCE

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| <b>Communication Theory (TA)</b><br><i>KAIST</i>                                                                                                                                                            | Feb. 2025 – June 2025<br><i>Daejeon, Korea</i> |
| <b>Undergraduate Internship Mentor</b><br><i>KAIST</i>                                                                                                                                                      | Sep. 2024 – Dec. 2024<br><i>Daejeon, Korea</i> |
| <ul style="list-style-type: none"><li>• Advised an undergraduate student's thesis research</li><li>• Research topic: joint channel and time-interleaved ADC mismatch compensation in OFDM systems</li></ul> |                                                |
| <b>Global Educational Program for Samsung Semiconductor (TA)</b><br><i>KAIST</i>                                                                                                                            | June 2024 – June 2025<br><i>Daejeon, Korea</i> |
| <b>Aramco Coding School</b><br><i>Saudi Arabian Oil Company</i>                                                                                                                                             | Aug. 2022 – Nov. 2022<br><i>Ulsan, Korea</i>   |
| <ul style="list-style-type: none"><li>• Developed an iOS application for protecting the environment with middle school students</li></ul>                                                                   |                                                |
| <b>A.I. 4.0 Studio</b><br><i>UNIST</i>                                                                                                                                                                      | May 2022 – Nov. 2022<br><i>Ulsan, Korea</i>    |
| <ul style="list-style-type: none"><li>• Implemented <i>pix2pix</i> (image-to-image translation with cGAN) with high school students using TensorFlow</li></ul>                                              |                                                |
| <b>Calculus I (TA)</b><br><i>UNIST</i>                                                                                                                                                                      | Mar. 2020 – June 2020<br><i>Ulsan, Korea</i>   |
| <b>Freshman English Camp (TA)</b><br><i>UNIST</i>                                                                                                                                                           | Jan. 2020<br><i>Ulsan, Korea</i>               |
| <b>Contemporary Philosophy (TA)</b><br><i>UNIST</i>                                                                                                                                                         | Sep. 2019 – Dec. 2019<br><i>Ulsan, Korea</i>   |
| <b>Ulju-gun Science Mentoring for the Gifted</b><br><i>UNIST</i>                                                                                                                                            | Jan. 2019<br><i>Ulsan, Korea</i>               |
| <ul style="list-style-type: none"><li>• Taught algebra for middle school students during a 3-week camp</li></ul>                                                                                            |                                                |